

① Rate

A **rate** is a comparison of two quantities of different kinds by division with a unit.

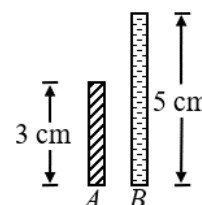
e.g. Dennis runs 8 m per second, then running speed = 8 m/s

② Ratio

A **ratio** is a comparison of two quantities of the same kinds by division without a unit.

A **ratio** of a quantity p to a quantity $q = p : q = \frac{p}{q}$, where $p \neq 0$ and $q \neq 0$.

e.g. Ratio of the lengths of A to $B = 3 \text{ cm} : 5 \text{ cm} = 3 : 5$



③ Properties of Ratios

1. If $k \neq 0$, then

$$(a) a : b = a \times k : b \times k \quad (b) a : b = \frac{a}{k} : \frac{b}{k}$$

2. A ratio $p : q$ is in its simplest form if both p and q are integers and their H.C.F. = 1

④ Three-term Ratios

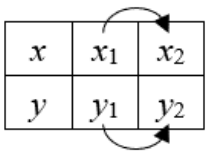
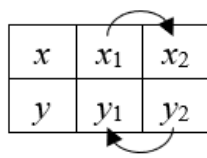
Consider $a : b : c = 5 : 2 : 4$. We have:

$$a : b = 5 : 2$$

$$b : c = 2 : 4 = 1 : 2 \quad (\text{proportion})$$

$$a : c = 5 : 4$$

⑤ Direct Proportions and Inverse Proportions

Direct Proportion	Inverse Proportion
$x_1 : x_2 = y_1 : y_2$ $\frac{x_1}{x_2} = \frac{y_1}{y_2}$ $\frac{x_1}{y_1} = \frac{x_2}{y_2}$	$x_1 : x_2 = y_2 : y_1$ $\frac{x_1}{x_2} = \frac{y_2}{y_1}$ $x_1 y_1 = x_2 y_2$
	

1. Johnny cycles 16 km in half an hour. Find the cycling speed of Johnny in km/h. (1 mark)

2. In a test, there were n students and 60 of them failed the test. The ratio of the number of students failed to the number of students passed is $5 : 4$. Find n . (1 mark)
3. In each of the following, x and y are non-zero numbers. Find $x : y$.
- (a) $3x = y$ (1 mark)
- (b) $\frac{x}{2} = \frac{y}{0.8}$ (1 mark)
4. The scale of a map is $1 : 500\,000$.
- (a) Represent the scale of the map in the form $1\text{ cm} : x\text{ km}$. (2 marks)
- (b) The distance between P and Q on the map is 3.6 cm . Find the actual distance between P and Q. (2 marks)
5. x and y are in direct proportion. Complete the table below. (2 marks)

x	3	12	
y	7		35

6. x and y are in inverse proportion. Complete the table below.

(2 marks)

x	30		12
y	4	6	

7. 6 boxes can hold products of total weight 240 kg. Jessie has 8 boxes. She claims that her boxes can hold products of total weight 300 kg. Do you agree? Explain your answer.

(2 marks)

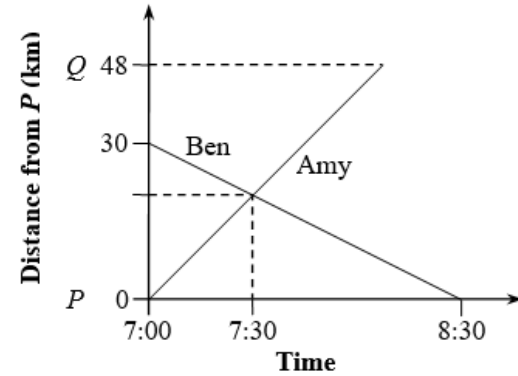
8. Paul walks from shop X to shop Y at an average speed of 6 km/h for 24 min and then he walks from shop Y to shop Z at an average speed of 7.5 km/h. It is given that he walks at an average speed of 110 m/min for the whole journey. How long does he take to walk from shop Y to shop Z? (4 marks)

9. In a hospital, the ratio of the number of male patients to the number of female patients is 3 : 2. If 20 male patients and 8 female patients are discharged, then the ratio of the number of male patients to the number of female patients is 5 : 4. Find the original number of male patients.

(3 marks)

10. The figure shows the graphs of Ben and Amy driving along a straight road PQ.

- (a) Find the speed of Ben in km/h. (1 mark)
- (b) Find the distance travelled by Ben when he meets Amy. (1 mark)
- (c) Find the speed of Amy in km/h. (1 mark)



11. On a bookshelf, there are Chinese books and English books only. The numbers of Chinese books and English books are in the ratio 9 : 4. It is given that there are 45 more Chinese books than the English books.

- (a) Find the numbers of Chinese books and English books on the bookshelf. (2 marks)
- (b) If 4 Chinese books and 3 English books are taken away, find the ratio of the number of Chinese books to the number of English books on the bookshelf now. (2 marks)

12. Given that $9y - 2x = 3x + 4y$ where x and y are non-zero numbers. Find $x : y$. (1 mark)

- 5

15. It is given that x , y and z are non-zero numbers such that $\frac{1}{x} : \frac{1}{y} : \frac{1}{z} = 4 : 5 : 3$.

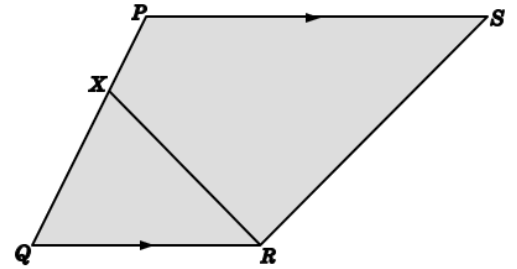
(a) Find $x : y : z$. (3 marks)

(b) Hence, find the value of $\frac{x+y}{y+z}$. (2 marks)

16. (a) On a map with scale $1 : 2000$, the dimensions of a soccer field are $5 \text{ cm} \times 3.2 \text{ cm}$. Find the actual area (in m^2) of the soccer field. (2 marks)
- (b) It is known that the cost of replacing the grass for an area of 20m^2 in the soccer field is \$2500 and 60% of the grass in the soccer field needs to be replaced. Joe claims that a budget of \$400000 is enough. Do you agree? Explain your answer. (2 marks)

16. Tea leaves A and tea leaves B are mixed in the ratio $x : y$ by weight. The costs of tea leaves A and tea leaves B are \$ 80/kg and \$ 110/kg respectively. If the cost of the mixed tea leaves is \$ 98/kg, find $x : y$. (2 marks)
17. Anna and Barry have some almonds. If Barry gives one-third of his almond to Anna, the numbers of almonds of Anna and Barry will be in the ratio 3 : 2. Find the original ratio. (2 marks)

18. In the figure, PQRS is a trapezium and $PS : QR = 3 : 2$. X is a point lying on PQ such that $PX : XQ = 3 : 5$. Find the ratio area of $\triangle XQR$: area of quadrilateral PXRS. (7 marks)



19. The bathtub in Chan's family has a hot water tap and a cold water tap. Below shows how Mrs Chan filled up the empty bathtub on Monday and Tuesday.

Monday: First, only turned on the hot water tap for 13 minutes. Then, only turned on the cold water tap for 8 minutes.

Tuesday: Only turned on the hot water tap for 25 minutes.

Suppose Mrs Chan tries to fill up the empty bathtub by turning on both taps at the same time on Wednesday. Find the time needed. (5 marks)